Routing, TCP/UDP and Application Protocols

Computer Networks - Units 3-4 - Network, transport and application layers

Routing

- A router forwards packets by longest-prefix match in its forwarding table.
- Distance-vector (RIP) shares distance tables with neighbours using Bellman-Ford; it can suffer count-to-infinity.
- Link-state (OSPF) floods link information so each router runs Dijkstra on the full topology.
- BGP exchanges routes between autonomous systems on the Internet.

UDP vs TCP

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Feature        UDP               TCP
Connection     none              3-way handshake
Reliability    none              ACKs, retransmission
Ordering       none              sequence numbers
Overhead      8-byte header     20+ byte header
Use            DNS, streaming    web, email, file transfer
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TCP connection and control

- Establish: SYN, SYN-ACK, ACK. Close: FIN/ACK exchange in each direction.
- Flow control uses the receiver window so a fast sender cannot overwhelm a slow receiver.
- Congestion control: slow start grows the window exponentially until a threshold, then congestion avoidance grows it linearly; on loss the window shrinks.

Application layer

- DNS translates names to IP addresses using a hierarchy of resolvers and name servers.
- HTTP is a request-response protocol; HTTPS is HTTP over TLS.
- SMTP sends mail; IMAP and POP3 retrieve it.

> *What happens when you type a URL? DNS lookup, TCP handshake, TLS handshake, HTTP request, server response, browser renders the page.*